# COMPACTNESS AND INERTIA, WITH SOLUTIONS

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Let $E$ be an elliptic curve over a local field $K$. The rational points of $E$ with nonsingular reduction form a subgroup $E_0(K)$ of finite index in $E(K)$ [Sil09, Theorem VII.6.1 and Corollary VII.6.2], and $E$ has good reduction if and only if the inertia group acts trivially on its torsion, by the criterion of Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]. Problem 1.8 proves the finiteness of $E(K)/E_0(K)$ by a compactness argument. Problem 2.12 studies how the inertia group acts on the torsion of a curve with potential good reduction. Each problem is preceded by the definitions and results it uses.

**Standing notation.**

- A finite extension $K$ of $\mathbb{Q}_p$, with normalized valuation $v$, ring of integers $\mathcal{O}_K$, uniformizer $\pi$ and residue field $k = \mathcal{O}_K/\pi\mathcal{O}_K$, a finite field of characteristic $p$. (Silverman's Chapter VII assumes that $K$ and $k$ are perfect, and here both are.)
- A fixed algebraic closure $\overline{K}$ of $K$.
- A bar denotes reduction modulo $\pi$.
- An elliptic curve $E$ over $K$, given by a minimal Weierstrass equation $F(X, Y, Z) = 0$, with $F \in \mathcal{O}_K[X, Y, Z]$ homogeneous of degree 3. It embeds $E$ in $\mathbb{P}^2$ as $V_+(F)$.
- The primitive coordinates $[X : Y : Z]$ of $P \in \mathbb{P}^2(K)$: coordinates in $\mathcal{O}_K$, not all in $\pi\mathcal{O}_K$. Every $P$ has them, and they are unique up to a factor in $\mathcal{O}_K^\times$.
- The reduction $\bar{P} \coloneqq [\bar{X} : \bar{Y} : \bar{Z}] \in \mathbb{P}^2(k)$ of $P \in \mathbb{P}^2(K)$, computed from primitive coordinates, hence well defined.
- The reduced curve $E_\pi$, defined by $\bar{F} = 0$. If $P \in E(K)$, then $\bar{P} \in E_\pi(k)$.

**How to use this.** Allow about 25 minutes for each problem. The `Magma` block in Problem 2.12 runs in seconds in the free online calculator. Everything else is by hand.

## 1. Finiteness by compactness

This section is [Sil09, Exercise 7.6]. The argument uses that $k$ is finite, so it does not apply, for instance, to the completion of the maximal unramified extension of $\mathbb{Q}_p$: a field complete with respect to a discrete valuation, with residue field $\overline{\mathbb{F}}_p$ (item (6)). Yet this is the case that the criterion of Néron–Ogg–Shafarevich needs: Silverman's proof of it uses the finiteness of $E(K^{\mathrm{unr}})/E_0(K^{\mathrm{unr}})$, where $K^{\mathrm{unr}}$ is the maximal unramified extension of $K$, whose residue field is an algebraic closure of $k$ [Sil09, proof of Theorem VII.7.1].

**Definition 1.1** (The topology of $K$)**.** We give $K$ the topology defined by $v$: the sets $a + \pi^n\mathcal{O}_K$, for $a \in K$ and $n \in \mathbb{Z}$, form a basis, and each of them is both open and closed. With it, $K$ is a Hausdorff topological field.

**Definition 1.2** (The subgroup $E_0(K)$)**.** Let $E_\pi^{\mathrm{ns}}$ be $E_\pi$ minus its singular point, if it has one. Then

$$E_0(K) \coloneqq \{P \in E(K) : \bar{P} \in E_\pi^{\mathrm{ns}}(k)\}.$$

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**Definition 1.3** (Regular functions [Poo06, Definition 1.8.3])**.** A function $\psi \in K(E)$ is regular at $Q \in E(\overline{K})$ if $\psi = g/h$ with $g, h \in K[X, Y, Z]$ homogeneous of the same degree and $h(Q) \neq 0$. Then $\psi(Q) \coloneqq g(Q)/h(Q)$, an element of $K$ if $Q \in E(K)$. The functions regular at $Q$ form a discrete valuation ring with fraction field $K(E)$.

**Proposition 1.4** ($E_0(K)$ is a subgroup [Sil09, Proposition VII.2.1])**.** *The set $E_0(K)$ is a subgroup of $E(K)$.*

**Proposition 1.5** (Compactness [Poo17, §1.1.2])**.** *The ring $\mathcal{O}_K$ is compact: it is complete, and for every $n$ it is covered by the $(\#k)^n$ translates of $\pi^n\mathcal{O}_K$. Equivalently, $K$ is locally compact.*

**Proposition 1.6** (Morphisms $E \to E$ over $K$)**.**

*(1) The morphisms $\alpha\colon E \to E$ defined over $K$ are exactly the points $\alpha = [\psi_0 : \psi_1 : \psi_2] \in E(K(E))$, that is, $\psi_0, \psi_1, \psi_2 \in K(E)$, not all zero, with $F(\psi_0, \psi_1, \psi_2) = 0$* [Poo17, Proposition 3.6.5(b)].
*(2) If $Q \in E(\overline{K})$ and $\psi_0, \psi_1, \psi_2$ are regular at $Q$ and not all zero there, then $\alpha(Q) = [\psi_0(Q) : \psi_1(Q) : \psi_2(Q)]$* [Sil09, §I.3].
*(3) Replacing each $\psi_i$ by $\phi\psi_i$, for $\phi \in K(E)^\times$, gives the same point of $E(K(E))$, and for every $Q \in E(\overline{K})$ some $\phi$ makes $\phi\psi_0, \phi\psi_1, \phi\psi_2$ regular at $Q$ and not all zero there: a power of a uniformizer of the discrete valuation ring of Definition 1.3 will do* [Sil09, proof of Proposition II.2.1].

**Proposition 1.7** (Translations [Sil09, Theorem III.3.6 and Example III.4.7])**.** *Let $P \in E(K)$. The translation $\tau_P\colon E \to E$, $Q \mapsto Q + P$, is an isomorphism defined over $K$, with inverse $\tau_{-P}$.*

**Problem 1.8** ($E(K)/E_0(K)$ is finite)**.**
Let $q\colon K^3 - \{0\} \to \mathbb{P}^2(K)$ be the quotient map. Give $K^3 - \{0\}$ the product topology and $\mathbb{P}^2(K)$ the quotient topology: $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U)$ is open. Let

$$S \coloneqq \{\mathbf{x} = (x_0, x_1, x_2) \in \mathcal{O}_K^3 : x_i \in \mathcal{O}_K^\times \text{ for some } i\}$$

be the set of primitive coordinate vectors.

(1) Show that $S$ is compact and open in $K^3 - \{0\}$, and that $q(S) = \mathbb{P}^2(K)$. Deduce that $\mathbb{P}^2(K)$ is compact. Show that $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U) \cap S$ is open in $S$.

Solution. $S = \mathcal{O}_K^3 - (\pi\mathcal{O}_K)^3$. By Proposition 1.5 and Tychonoff's theorem, $\mathcal{O}_K^3$ is compact, and $(\pi\mathcal{O}_K)^3$ is open, so $S$ is closed in a compact space, hence compact. It is also open: $S$ is the union over $i$ of the sets $\mathcal{O}_K \times \mathcal{O}_K \times \mathcal{O}_K$ with the $i$-th factor replaced by $\mathcal{O}_K^\times$, and $\mathcal{O}_K^\times$ is open, being the union of the balls $u + \pi\mathcal{O}_K$ with $u$ a unit. If $\mathbf{x} \in K^3 - \{0\}$ and $m = \min_i v(x_i)$, then $\pi^{-m}\mathbf{x} \in S$ and $q(\pi^{-m}\mathbf{x}) = q(\mathbf{x})$, so $q(S) = \mathbb{P}^2(K)$. As $q$ is continuous, $\mathbb{P}^2(K) = q(S)$ is compact.
If $U$ is open, then $q^{-1}(U) \cap S$ is open in $S$. Conversely, suppose that $V \coloneqq q^{-1}(U) \cap S$ is open in $S$, hence in $K^3 - \{0\}$, since $S$ is open. Every $\mathbf{x} \in K^3 - \{0\}$ is $c\,\mathbf{s}$ with $c \in K^\times$ and $\mathbf{s} \in S$, and $q^{-1}(U)$ is stable under $K^\times$, so

$$q^{-1}(U) = \bigcup_{c \in K^\times} cV.$$

Multiplication by $c$ is a homeomorphism of $K^3 - \{0\}$, so this is a union of open sets, and $U$ is open.

(2) Show that $E(K) = V_+(F)(K)$ is closed in $\mathbb{P}^2(K)$, and deduce that it is compact.

Solution. The set $q^{-1}(E(K)) = \{\mathbf{x} \in K^3 - \{0\} : F(\mathbf{x}) = 0\}$ is closed, because $F$ is a continuous function $K^3 \to K$ ($K$ is a topological field) and $\{0\}$ is closed in the Hausdorff space $K$. So $E(K)$ is closed in the compact space $\mathbb{P}^2(K)$, hence compact.

(3) Let $g, h \in K[X, Y, Z]$ be homogeneous of the same degree. Show that $W := \{Q \in \mathbb{P}^2(K) : h(Q) \neq 0\}$ is open, and that $W \to K$, $Q \mapsto g(Q)/h(Q)$, is continuous. Deduce from Propositions 1.6 and 1.7 that every translation $\tau_P \colon E(K) \to E(K)$ is a homeomorphism.

Solution. As in (2), the set where $h$ vanishes is closed, so $W$ is open. The function $G(\mathbf{x}) := g(\mathbf{x})/h(\mathbf{x})$ is continuous on the open set $q^{-1}(W)$, and $G(c\mathbf{x}) = G(\mathbf{x})$ for $c \in K^\times$, since $g$ and $h$ have the same degree. So $G = \theta \circ q$ for a function $\theta \colon W \to K$. If $V \subseteq K$ is open, then $q^{-1}(\theta^{-1}(V)) = G^{-1}(V)$ is open in $q^{-1}(W)$, which is open in $K^3 - \{0\}$. So $\theta^{-1}(V)$ is open, and $\theta$ is continuous.
Let $P, Q \in E(K)$. By Proposition 1.7, $\tau_P$ is a morphism defined over $K$. Write $\tau_P = [\psi_0 : \psi_1 : \psi_2]$ as in Proposition 1.6, with the $\psi_i$ regular at $Q$ and $\psi_j(Q) \neq 0$ for some $j$ (part (3)). Write $\psi_i = g_i/h_i$ with $h_i(Q) \neq 0$ (Definition 1.3). On the open neighborhood $U := E(K) \cap \{h_0 h_1 h_2 \neq 0\}$ of $Q$ in $E(K)$, each $\psi_i$ is regular and, by the first part, continuous. So $U' := \{Q' \in U : \psi_j(Q') \neq 0\}$ is an open neighborhood of $Q$. On $U'$, by part (2), $\tau_P$ is the composition of the continuous maps

$$U' \to K^3 - \{0\},\ Q' \mapsto \big(\psi_0(Q'), \psi_1(Q'), \psi_2(Q')\big), \qquad \text{and} \qquad q \colon K^3 - \{0\} \to \mathbb{P}^2(K).$$

So $\tau_P$ is continuous on each set of an open cover of $E(K)$, hence continuous. Its inverse $\tau_{-P}$ is continuous for the same reason, so $\tau_P$ is a homeomorphism.

(4) Show that the reduction map $\mathbb{P}^2(K) \to \mathbb{P}^2(k)$, $P \mapsto \bar{P}$, is locally constant. Deduce that $E_0(K)$ is open in $E(K)$, and also closed.

Solution. For $\mathbf{x} \in S$, the open neighborhood $\mathbf{x} + (\pi\mathcal{O}_K)^3$ of $\mathbf{x}$ lies in $S$, and every $\mathbf{x}'$ in it has $\bar{\mathbf{x}}' = \bar{\mathbf{x}}$. So $\mathbf{x} \mapsto \overline{q(\mathbf{x})}$ is locally constant on $S$. Hence for every subset $T \subseteq \mathbb{P}^2(k)$, the set $q^{-1}(\{P : \bar{P} \in T\}) \cap S$ is a union of such neighborhoods, so open in $S$, and $\{P : \bar{P} \in T\}$ is open by (1). Its complement is the set for $\mathbb{P}^2(k) - T$, so it is also closed. With $T = E_\pi^{\mathrm{ns}}(k)$, $E_0(K)$ is the intersection of $E(K)$ with an open and closed subset of $\mathbb{P}^2(K)$.

(5) Prove that $E(K)/E_0(K)$ is finite.

Solution. By Proposition 1.4, the cosets $P + E_0(K) = \tau_P(E_0(K))$ partition $E(K)$, and each is open by (3) and (4). By (2), $E(K)$ is compact, so finitely many cosets cover it. Since the cosets are disjoint and nonempty, each one must appear in this finite subcover, so there are finitely many.

(6) Let $K'$ be a field complete with respect to a discrete valuation whose residue field $k'$ is infinite, for instance algebraically closed. Show that its ring of integers is not compact. Which step of the argument above fails for an elliptic curve over $K'$?

Solution. **The first step, (1): $\mathbb{P}^2(K')$ is not compact.** Choose lifts $a_\alpha \in \mathcal{O}_{K'}$ of the elements $\alpha \in k'$. The balls $a_\alpha + \pi\mathcal{O}_{K'}$ are open, pairwise disjoint and cover $\mathcal{O}_{K'}$, and there are infinitely many of them. Since none can be removed, $\mathcal{O}_{K'}$ is not compact. So the first step fails: $\mathcal{O}_{K'}^3$, $S$ and $\mathbb{P}^2(K')$ are not compact, and for an elliptic curve $E'$ over $K'$, the cover of $E'(K')$ by the cosets of $E_0'(K')$ need have no finite subcover.

*Remark.* The finiteness of $E'(K')/E_0'(K')$ is still true, but it needs the theory of Néron models [Sil09, Corollary VII.6.2].
Compare [Poo17, §2.6.2 and Proposition 2.6.1], which define the same kind of topology on $X(K)$ for every variety $X$ over $K$, and show that $X(K)$ is compact when $X$ is proper and $K$ is a local field.

## 2. Inertia acting on torsion

This section is [Sil09, Exercise 7.9], with the hypothesis $m \geqslant 3$ added to its first two parts (item (5) explains why) and a `Magma` check. We use the notation of [Sil09, §VII.4].

**Definition 2.1** (Reduction types)**.** The curve $E$ has good reduction if $E_\pi$ is nonsingular, that is, if $v(\Delta) = 0$, and bad reduction otherwise. It has potential good reduction if it has good reduction over some finite extension of $K$.

**Definition 2.2** (Inertia)**.** Let $\mathrm{Gal}_K := \mathrm{Gal}(\overline{K}/K)$, $K^{\mathrm{unr}} \subseteq \overline{K}$ the maximal unramified extension of $K$, and $I_v := \mathrm{Gal}(\overline{K}/K^{\mathrm{unr}})$ the inertia group. A set on which $\mathrm{Gal}_K$ acts is unramified if $I_v$ acts trivially on it.

**Definition 2.3** (Ramification of finite extensions)**.** For an extension $L$ of $K$ inside $\overline{K}$, let $\mathrm{Gal}_L := \mathrm{Gal}(\overline{K}/L)$. If $L/K$ is finite and Galois, the inertia group of $L/K$ is the image of $I_v$ in $\mathrm{Gal}(L/K)$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_L)$. The extension $L/K$ is unramified if its inertia group is trivial, and tamely ramified if its order is prime to $p$.

**Definition 2.4** (Torsion fields and Tate modules [Sil09, §III.7])**.** For $m \geqslant 1$ with $p \nmid m$, $K(E[m])$ is the field generated over $K$ by the coordinates of the points of $E[m](\overline{K})$, so $\mathrm{Gal}_{K(E[m])}$ is the kernel of the action of $\mathrm{Gal}_K$ on $E[m](\overline{K})$. For a prime $\ell \neq p$, the Tate module $T_\ell(E)$ is the inverse limit of the $E[\ell^n](\overline{K})$ along the surjections $[\ell]$, and $\rho_\ell \colon \mathrm{Gal}_K \to \mathrm{Aut}\, T_\ell(E)$ is the action of $\mathrm{Gal}_K$ on it.

In the following results, $\ell \neq p$ is a prime and $m \geqslant 1$ is an integer with $p \nmid m$.

**Proposition 2.5** (Minimal equations [Sil09, Remark VII.1.1 and §III.1])**.** *A Weierstrass equation with coefficients in $\mathcal{O}_K$ and $v(\Delta) < 12$ is minimal. For $y^2 = x^3 + Ax + B$, $\Delta = -16(4A^3 + 27B^2)$.*

**Corollary 2.6** (Torsion [Sil09, Corollary III.6.4(b)])**.** *The group $E[m](\overline{K})$ is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$.*

**Proposition 2.7** (The Tate module [Sil09, Proposition III.7.1(a)])**.** *The Tate module $T_\ell(E)$ is isomorphic to $\mathbb{Z}_\ell^2$. Hence each projection $T_\ell(E) \to E[\ell^n](\overline{K})$ is a surjective homomorphism of $\mathrm{Gal}_K$-modules, and its kernel is $\ell^n T_\ell(E)$, since both have index $\ell^{2n}$.*

**Proposition 2.8** (Good reduction [Sil09, Proposition VII.4.1])**.** *If $E$ has good reduction, then $E[m](\overline{K})$ and $T_\ell(E)$ are unramified.*

**Theorem 2.9** (Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1])**.** *If $T_\ell(E)$ is unramified, then $E$ has good reduction.*

**Corollary 2.10** (Potential good reduction [Sil09, Corollary VII.7.3])**.** *The curve $E$ has potential good reduction if and only if $\rho_\ell(I_v)$ is finite.*

**Proposition 2.11** (Inertia groups of finite extensions [Sil09, proof of Corollary VII.7.3])**.** *If $H \subseteq I_v$ is a closed normal subgroup of finite index, there is a finite extension $K'$ of $K$ inside $\overline{K}$ whose inertia group $\mathrm{Gal}(\overline{K}/K'K^{\mathrm{unr}}) = I_v \cap \mathrm{Gal}_{K'}$ is $H$.*

**Problem 2.12** (The inertia group of $K(E[m])/K$)**.**
Suppose that $E$ has potential good reduction, and let $J$ be the subgroup of the elements of $I_v$ that act trivially on $E[m](\overline{K})$ for every $m \geqslant 1$ with $p \nmid m$.

(1) Let $\ell$ be a prime and $r \geqslant 1$ with $\ell^r \geqslant 3$, and let $g \in \mathrm{GL}_2(\mathbb{Z}_\ell)$ be an element of finite order with $g \equiv I \pmod{\ell^r}$. Show that $g = I$.

Solution. Suppose that $g \neq I$. Some power of $g$ has prime order $q$, and it is still congruent to $I$ modulo $\ell^r$, so we may assume that $g$ has order $q$. Write $g = I + \ell^s h$ with $h \in M_2(\mathbb{Z}_\ell)$ not

divisible by $\ell$. Then $s \geqslant r$, so $\ell^s \geqslant 3$. Expanding,

$$0 = g^q - I = q\ell^s h + \sum_{i=2}^{q} \binom{q}{i} \ell^{is} h^i.$$

If $q \neq \ell$, every term of the sum is divisible by $\ell^{2s}$, so $q\ell^s h \equiv 0 \pmod{\ell^{s+1}}$ and, $q$ being a unit, $h \equiv 0 \pmod{\ell}$: a contradiction. If $q = \ell$, then $\ell$ divides $\binom{\ell}{i}$ for $2 \leqslant i < \ell$, so those terms are divisible by $\ell^{2s+1}$, hence by $\ell^{s+2}$. The last term is divisible by $\ell^{\ell s}$, and $\ell s \geqslant s+2$ because $s(\ell-1) \geqslant 2$: this is where $\ell^s \geqslant 3$ is needed. So $\ell^{s+1} h \equiv 0 \pmod{\ell^{s+2}}$, and again $h \equiv 0 \pmod{\ell}$.

*Remark.* For $\ell^r = 2$ the statement is false: $g = -I$ has order 2 and $-I \equiv I \pmod 2$.

(2) Show that $J = I_v \cap \ker \rho_\ell$ for every prime $\ell \neq p$, and that $J$ has finite index in $I_v$.

Solution. Let $J_\ell := I_v \cap \ker \rho_\ell$. An element of $J$ acts trivially on every $E[\ell^n](\overline{K})$, hence on their inverse limit $T_\ell(E)$, so $J \subseteq J_\ell$. Conversely, $\rho_\ell(I_v)$ is finite by Corollary 2.10, so $J_\ell$, the kernel of $I_v \to \rho_\ell(I_v)$, has finite index. It is normal, and closed, being the intersection of the kernels of the actions on the finite sets $E[\ell^n](\overline{K})$. By Proposition 2.11, $J_\ell$ is the inertia group of a finite extension $K'$ of $K$, again a finite extension of $\mathbb{Q}_p$. Over $K'$, $T_\ell(E)$ is unramified, so $E$ has good reduction over $K'$ (Theorem 2.9). Then $E[m](\overline{K})$ is unramified over $K'$ for every $m$ with $p \nmid m$ (Proposition 2.8), that is, $J_\ell$ acts trivially on it. So $J_\ell \subseteq J$.

(3) Let $m \geqslant 3$ with $p \nmid m$. Show that $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and deduce that the inertia group of $K(E[m])/K$ is isomorphic to $I_v/J$, independently of $m$.

Solution. By definition, $J \subseteq I_v \cap \mathrm{Gal}_{K(E[m])}$. Conversely, since $m \geqslant 3$, it has a divisor $\ell^r \geqslant 3$ with $\ell$ prime: an odd prime factor of $m$, or 4 if $m$ is a power of 2. Then $\ell \neq p$, as $p \nmid m$, and $E[\ell^r] \subseteq E[m]$. Let $\sigma \in I_v \cap \mathrm{Gal}_{K(E[m])}$. Then $\sigma$ acts trivially on $E[\ell^r](\overline{K}) \cong T_\ell(E)/\ell^r T_\ell(E)$ (Proposition 2.7), so the matrix of $\rho_\ell(\sigma)$ in a $\mathbb{Z}_\ell$-basis of $T_\ell(E)$ is congruent to $I$ modulo $\ell^r$. It has finite order, since $\rho_\ell(I_v)$ is finite (Corollary 2.10), so $\rho_\ell(\sigma) = I$ by (1), and $\sigma \in J$ by (2). So $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and the inertia group of $K(E[m])/K$ is isomorphic to $I_v/(I_v \cap \mathrm{Gal}_{K(E[m])}) = I_v/J$.

(4) Let $m \geqslant 3$ with $p \nmid m$. Show that $K(E[m])/K$ is unramified if and only if $E$ has good reduction. Which direction holds for every $m \geqslant 1$ with $p \nmid m$, and for every elliptic curve over $K$?

Solution. **Good reduction implies unramified, for all such $m$ and $E$.** If $E$ has good reduction, then $E[m](\overline{K})$ is unramified by Proposition 2.8, so $K(E[m])/K$ is unramified. This holds for every $m$ with $p \nmid m$ and every elliptic curve over $K$. Conversely, suppose that $m \geqslant 3$ and that $K(E[m])/K$ is unramified. Then $I_v/J$ is trivial by (3), so $I_v = J = I_v \cap \ker \rho_\ell$ for a prime $\ell \neq p$ by (2). So $T_\ell(E)$ is unramified, and $E$ has good reduction by Theorem 2.9.

(5) Let $p \geqslant 3$ and $E\colon y^2 = x^3 - \pi^2 x$.

(a) Show that $E$ has bad reduction, potential good reduction, and $K(E[2]) = K$.

Solution. Here

$$\Delta = -16 \cdot 4(-\pi^2)^3 = 64\pi^6, \qquad v(\Delta) = 6 < 12,$$

since $p$ is odd. So the equation is minimal (Proposition 2.5) and $E$ has bad reduction. Over $K(\sqrt{\pi})$, $E$ is isomorphic to $E_1\colon y_1^2 = x_1^3 - x_1$, as shown in (5)(b), and $E_1$ has $\Delta = 64$, a unit: $E$ has potential good reduction. And $x^3 - \pi^2 x = x(x-\pi)(x+\pi)$, so the points of order two $(0,0)$, $(\pm\pi, 0)$ are rational, and $K(E[2]) = K$.

*Remark.* Compare: $j(E) = 1728 \in \mathcal{O}_K$, and [Sil09, Proposition VII.5.5].

(b) Show that $I_v/J$ has order 2, and that its nontrivial element acts on every $T_\ell(E)$ as $-1$.

SOLUTION. Let $E_1\colon y_1^2 = x_1^3 - x_1$, with $\Delta = 64$, a unit: it has good reduction over every finite extension of $K$. Fix a square root $\sqrt{\pi} \in \overline{K}$. The map $\phi\colon E_1 \to E$, $(x_1, y_1) \mapsto (\pi x_1, \pi\sqrt{\pi}\, y_1)$, is an isomorphism, since

$$(\pi\sqrt{\pi}\, y_1)^2 = \pi^3(x_1^3 - x_1) = (\pi x_1)^3 - \pi^2(\pi x_1).$$

Let $\sigma \in \mathrm{Gal}_K$ and $\chi(\sigma) \coloneqq \sigma(\sqrt{\pi})/\sqrt{\pi} \in \{\pm 1\}$. Then $\sigma(\phi(P)) = \phi(\chi(\sigma)\,\sigma(P))$ for $P \in E_1(\overline{K})$, because $(x_1, y_1) \mapsto (x_1, -y_1)$ is $[-1]$. If $\sigma \in I_v$, then $\sigma$ fixes $E_1[m](\overline{K})$ for every $m$ with $p \nmid m$ (Proposition 2.8), so $\sigma$ acts on $E[m](\overline{K}) = \phi(E_1[m](\overline{K}))$, and on every $T_\ell(E)$, as multiplication by $\chi(\sigma)$. Since $t^2 - \pi$ is Eisenstein, $K(\sqrt{\pi})/K$ is ramified, so $\chi$ is nontrivial on $I_v$. Hence $J = I_v \cap \ker\chi$ has index 2, and the other coset acts as $-1$.

(c) So (3) and (4) fail for $m = 2$. Where did your proofs use $m \geqslant 3$?

SOLUTION. **In (3), to find a prime power $\ell^r \geqslant 3$ dividing $m$.** For $m = 2$, $K(E[2]) = K$, so $I_v \cap \mathrm{Gal}_{K(E[2])} = I_v \neq J$, and (3) fails. And $K(E[2])/K$ is unramified although $E$ has bad reduction, so (4) fails. The proof of (3) needed a prime power $\ell^r \geqslant 3$ dividing $m$, which does not exist for $m = 2$: the element $-I \in \mathrm{GL}_2(\mathbb{Z}_2)$ has order 2 and is congruent to $I$ modulo 2, but not modulo 4.

*Remark.* This is why Silverman's hint works modulo 4 when $\ell = 2$.

(6) Suppose that $p \geqslant 5$. Show that $K(E[m])/K$ is tamely ramified for every $m \geqslant 1$ with $p \nmid m$.

SOLUTION. Since $p \geqslant 5$, (3) applies with $m = 3$: $I_v/J$ is isomorphic to the inertia group of $K(E[3])/K$, a subgroup of $\mathrm{Gal}(K(E[3])/K)$, which acts faithfully on $E[3](\overline{K}) \cong (\mathbb{Z}/3\mathbb{Z})^2$ (Corollary 2.6). So $I_v/J$ embeds in $\mathrm{GL}_2(\mathbb{F}_3)$, of order $48 = 2^4 \cdot 3$, and its order is prime to $p$. For every $m$ with $p \nmid m$, $J$ acts trivially on $E[m](\overline{K})$, so $J \subseteq I_v \cap \mathrm{Gal}_{K(E[m])}$, and the inertia group of $K(E[m])/K$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_{K(E[m])})$, is a quotient of $I_v/J$. So its order is prime to $p$ too.

(7) `Magma` computes $\#(I_v/J)$ for elliptic curves over $\mathbb{Q}_7$:

```
p := 7;
Inertia := func< A, B |
  #InertiaGroup(GaloisRepresentation(EllipticCurve([A, B]), p)) >;
Inertia(-1, 0);                    // 1
Inertia(-p^2, 0);                  // 2
Inertia(0, p);                     // 6
```

Check the first two values against your answers, and the third against (6).

SOLUTION. **They agree.** The first curve has $\Delta = 64$, a unit at 7: good reduction, so $I_v = J$ by (4) and (3). The second is the curve of (5) with $\pi = 7$, and $\#(I_v/J) = 2$. The third, $y^2 = x^3 + 7$, has $\#(I_v/J) = 6$, which divides 48. It acquires good reduction over $L = \mathbb{Q}_7(\varpi)$ with $\varpi^6 = 7$: the substitution $x = \varpi^2 x_1$, $y = \varpi^3 y_1$ gives $y_1^2 = x_1^3 + 1$. As $\mu_6(\overline{\mathbb{Q}}_7) \subseteq \mathbb{Q}_7$, $L/\mathbb{Q}_7$ is Galois, and it is totally ramified of degree 6. Its inertia group $I_v \cap \mathrm{Gal}_L$ lies in $J$, so $I_v/J$ is a quotient of the inertia group of $L/\mathbb{Q}_7$, of order 6. `Magma` says that it is all of it.

## REFERENCES


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